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DSE6111
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ABC Corporation Attrition Risk Final Report

Executive Summary

ABC Corporation's Data Science team conducted a data-driven analysis to understand and predict customer attrition, leveraging demographic, financial, and behavioral data to identify at-risk clients. Key variables included transaction history, card utilization, account type, relationship duration, and customer engagement metrics. By comparing trends in closed and active accounts, insights were gathered to target customers at risk of leaving.

Analysis and Findings:

The analysis utilized multiple predictive models, including Logistic Regression, K-Nearest Neighbors (KNN), Naive Bayes, Random Forest, and XGBoost. The XG-boost model provided feature importance that was in line with feature importance determined by the Logistic Regression model even though it appears to still be overfitted currently. Random forest also showed signs of good performance even though it is also currently overfitted. XG-boost may be the most important model to work on refining as the size of available data grows even though logistic regression is the most effective model at the moment. Naive Bayes and KNN showed promising results for the current datasets but may be less effective as the project evolves.Features of importance to focus on as determined by the logistic regression model and the XG-Boost Model (Exhibit C & Exhibit H) are customers with the highest total transaction count, highest total revolving balances, highest total relationship counts, and highest total changes between Q4 and Q1.

Recommendations:

ABC Corporation should focus on building customer loyalty by developing rewards programs and marketing the benefits of higher-tier cards to encourage continued engagement. Proactively enhancing customer engagement through regular, positive communication, not strictly limited to at-risk accounts, can strengthen relationships and foster trust. Additionally, offering financial

education programs can help at-risk customers improve their financial management, reducing defaults and benefiting both the customer and the company. By leveraging predictive insights, ABC Corporation can implement targeted interventions to address specific customer needs and allocate resources more effectively, ensuring a greater impact on retention efforts. Implementing these recommendations and leveraging additional data through a refined XG-boost model can give ABC Corporation the ability to retain customers, strengthen relationships, and improve outcomes for both stakeholders and clients.

Detailed Customer Attrition Report

Investigating the factors that impact a client's risk of attrition has proven itself to be a necessary step for ABC Corporation in order to retain valuable stakeholders of our organization and to improve the outcome of future consumers of our products. During difficult economic circumstances, maintaining the profitability of ABC Corporation through data-driven decision making is imperative while also providing impactful and strategic outreach to stakeholders who may be going through difficult financial times. These efforts can prove themselves to be a wise use of resources if they are approached with actions backed by insights that are statistically sound. Exploring our existing data through the methods of multiple predictive models has resulted in insights that can be used to take initial actions that can further be analyzed and improved upon to ensure that the growth of ABC Corporation is not as greatly affected by the current state of our economy. These efforts will also prove to be valuable as the market recovers for both consumers and corporations.

Approach and Data

The data collected about the consumers include valuable demographic information such as age, gender, level of education, number of dependents, marital status, and annual income. Information about total transaction amounts, number of transactions in the past twelve months, and average card utilization ratios can allow ABC Corporation to target consumers specifically who are deemed at risk of attrition by the models used to gather insights. This information can further be enhanced by observing variables such as the change in transaction amounts grouped by quarters, amount of months intactive in the past twelve months, total number of products held by the customer, and the amount of times contacted in the past twelve months. The type of card held by the customer, Blue, Silver, Gold, or Platinum, can also be taken into consideration as well as the period of the relationship with the bank and the average card utilization ratios. All of the aforementioned variables will be included as features in the supervised learning model with the exception of the customer number to weigh their impact against the variable of "attrition_flag" which measures closed versus active accounts. Determining trends among closed accounts and then comparing them to states of current active accounts will provide ABC Corporation with more accurate information needed to target at-risk clients given similar behaviors of former clients who now have closed accounts.

Data quality is a major factor when analyzing information that is available in order to make business decisions based on historic predictors. The ability to accurately predict behaviors of future actions relies strongly on the validity of the data and the soundness of the methods enacted to develop the predictive model. Preparing data for exploratory analysis through processes such as aggregation and advanced calculations involves removing duplicate data, resolving incomplete values, and coding qualitative variables so that their impacts may be taken into consideration. Balancing data points and subsetting the data into training sets and test sets also needs to be performed when assessing if the data is being skewed by inputs that do not accurately represent the population reflected within our data. By properly cleaning the data, more accurate and valuable insights can be obtained. Grouping variables to create new fields in the future can also provide actionable insights that improve upon the actions recommended after implementation.

Findings and Evaluation

The models used in the final exploration steps of our currently available data were Logistic Regression, K-nearest Neighbors, Naive Bates, Random Forest, and a final boosted model known as XG-boost. Each of these models have their strengths and weaknesses in regards to their ability to properly classify the information available to us. Logistic Regression models work well for interpretable, straightforward tasks like binary classification. K-nearest Neighbors models perform better on smaller datasets or simple problems and can be revisited for future comparisons if necessary. The Naive Bayes predictive model is highly efficient, easy to implement, and performs well on large datasets with independent features. The Random Forest model also handles large datasets and complex feature interactions effectively with high accuracy while reducing overfitting. In real world applications such as that with the availability of large datasets to ABC Corporation, XGBoost is ideal once over-fitting has been addressed.

Both XG-Boost and Random Forest unfortunately produced results that seemed highly accurate when tested with balanced data, but upon closer inspection proved to show more signs that the data was over-fitted which nullified the accuracy of the results calculated. Naive Bayes performed better than the null model when predicting classification of classes in the test set versus the training sets. K-Nearest Neighbors showed some improvement over the null model as the parameters were adjusted, but ultimately, of all the models explored, Logistic Regression performed the best with 90% accuracy in comparison to 84% accuracy in the Null model. Exact figures are presented in Exhibit A with the calculated plots also presented in the Appendix. The features of importance to focus on as determined by the logistic regression model and the XG-Boost Model (Exhibit B & Exhibit G) are customers with the highest total transaction count, highest total revolving balances, highest total relationship counts, and highest total changes between Q4 and Q1.

Recommendations

Customers have the option to spend their funds where they see fit, but the amount of funds available have lessened in the current environment. Building brand loyalty is one of the best strategies to improve the odds that consumer spending stays with ABC Corporation instead of these available funds being funneled to other businesses. Creating incentives for our client base to choose ABC Corporation over other alternatives is one of the best ways to keep stakeholders excited about our services. Also building relationships with these consumers is imperative in today's market. Increased contact and outreach should not be the action taken only when the client is at risk of unfortunate outcomes. These strategies should be part of actions taken when account status is favorable and when there are opportunities to increase the favorable opinion of ABC Corporation in the eyes of these consumers. Increasing marketing about the benefits of the different tiers of our cards to give the consumers exciting goals to strive for is one way that ABC Corporation can reduce the risk of attrition. Consumers often have several options when it comes to deciding which card to use for purchases, and building positive relationships with responsive and timely customer service can increase the odds for more favorable outcomes.

Developing incentives that outweigh the perceived benefit of other brands over that of ABC Corporation can help retain a consumer base that has the available funds to make more purchases. For those consumers who are at risk of attrition due to defaulting on payments, educational programs can be taken advantage of to improve the outcomes for both the customer and for ABC Corporation. Financial education is not always something that consumers have been exposed to at a level that is accessible and also attention keeping. Focusing efforts on these aspects as well should produce positive returns on investments made by ABC Corporation.

Conclusions

Strategically increasing communications and marketing efforts to customers who are at risk of attrition can only improve the outcomes for both the consumers and ABC Corporation. It is imperative to undertake this task by acting upon insights that are backed by statistically sound data analysis. By fitting the appropriate model to the data currently available to ABC Corporation and developing insights through further data collection, ABC Corporation can more accurately direct resources and actions to bring forward better results for both internal stakeholders and the customer base.

Appendix

Exhibit A: Summary of Model Accuracy and AUC

Model	**Accuracy**	**AUC**
Null Model	0.8405	N/A
Logistic Regression	0.905	0.9237
K-Nearest Neighbors	0.8765	0.825
Naive Bayes	0.8849	0.863
Random Forest	0.952	0.983
XG-Boost	N/A Due to Overfitting	0.992

Notes:

Multiple strategies were tested to address overfitting with Random Forest and XG-boost through adjusting parameters and engines. Ultimately the models produced the same outcome even when also rebalancing the data through ROSE.

Exhibit B: ROC and AUC for Logistic Regression Model

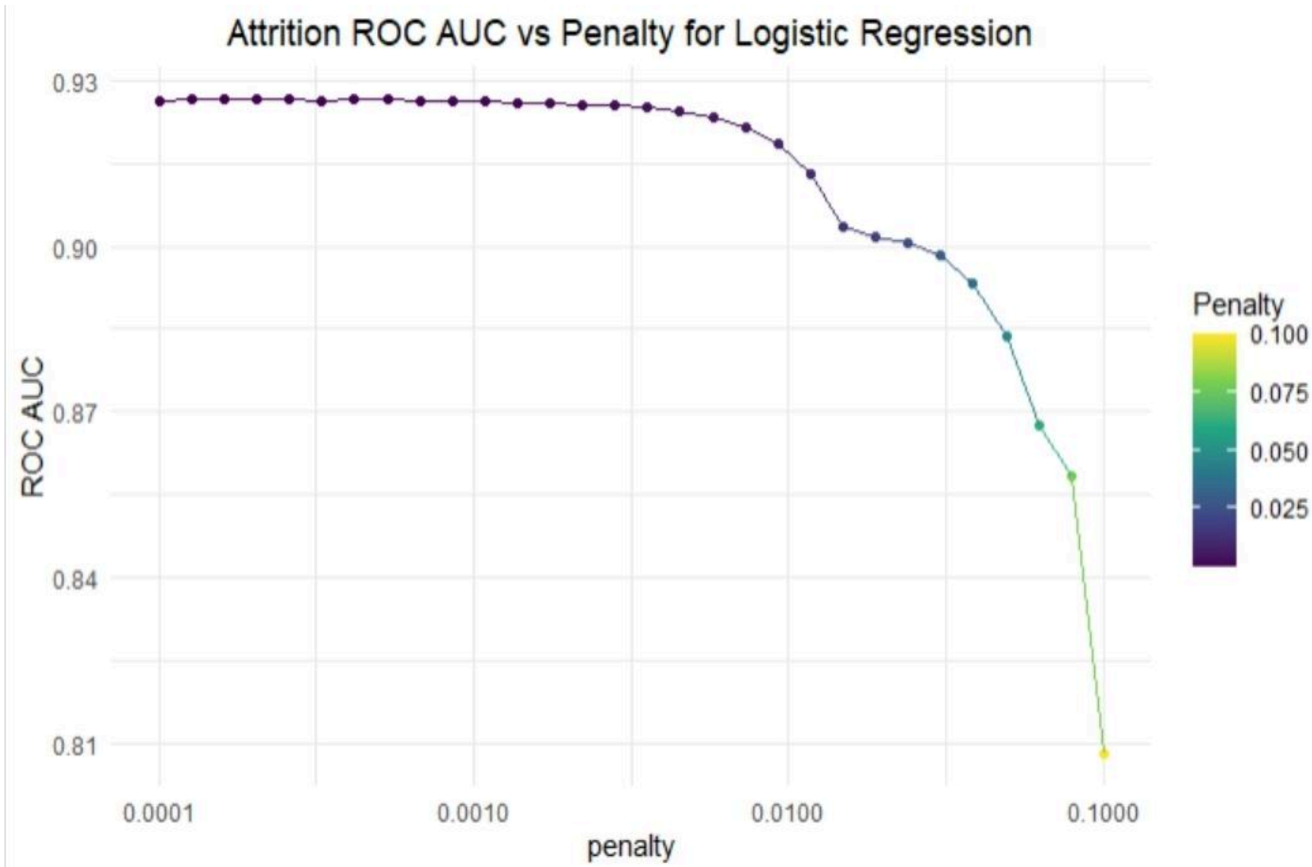

Exhibit C: Feature Importance for Logistic Regression Model

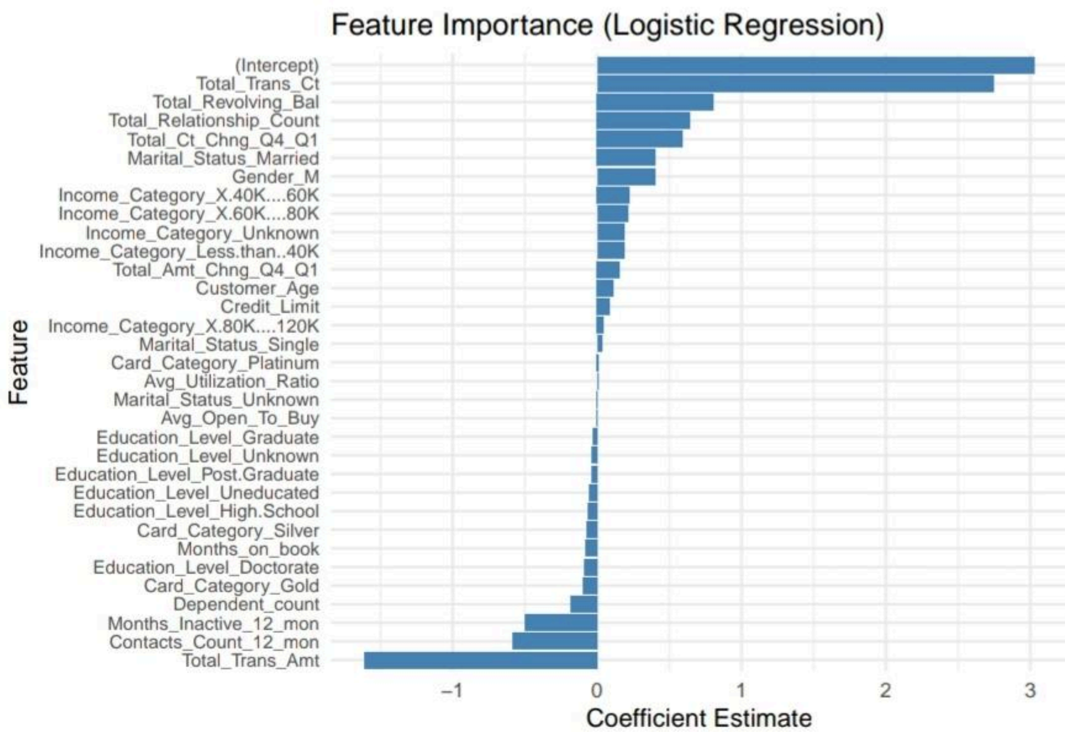

Exhibit D: ROC for K-nearest Neighbors Model

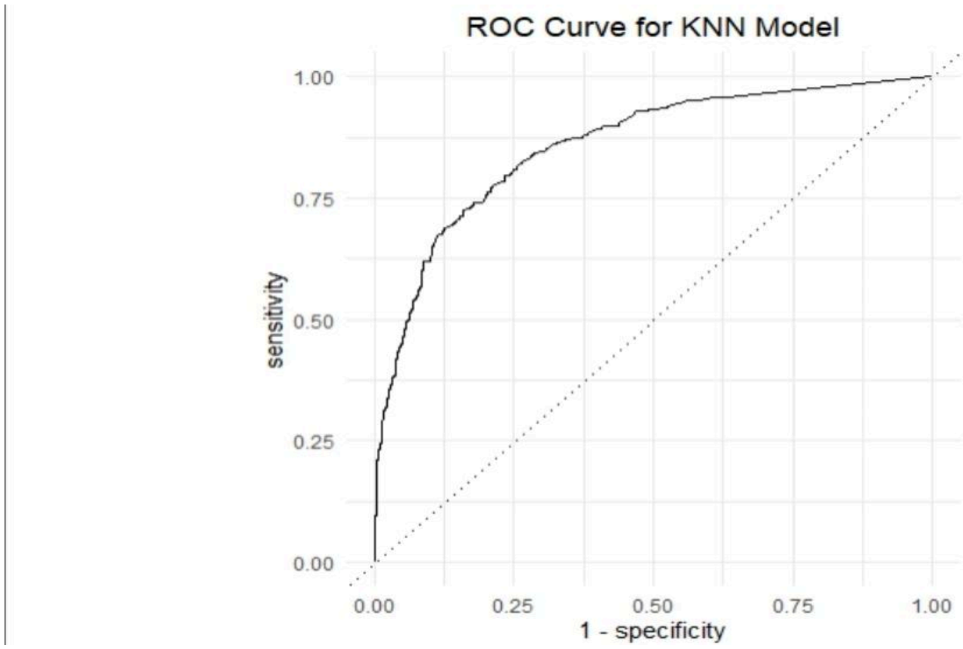

Exhibit E: ROC and AUC for Random Forest Model

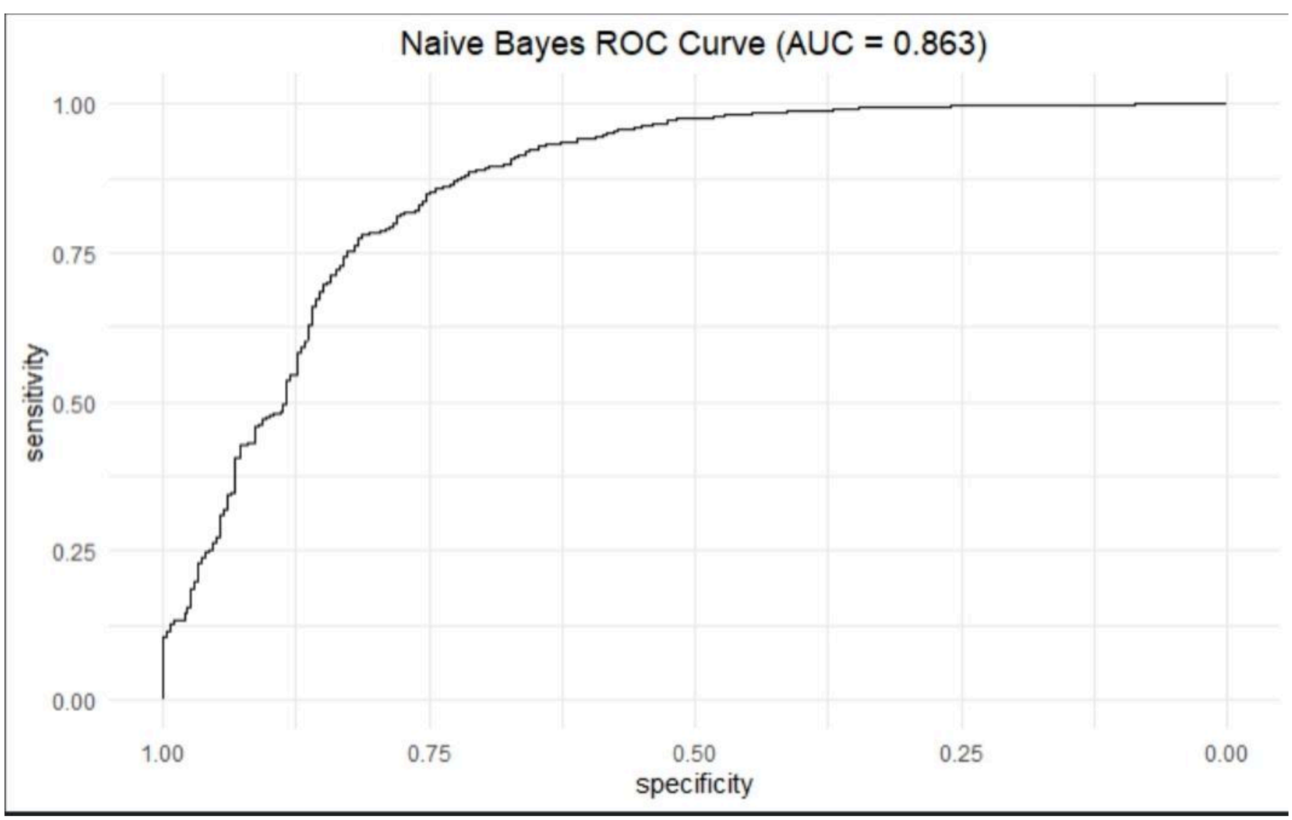

Exhibit F: ROC and AUC for Random Forest Model

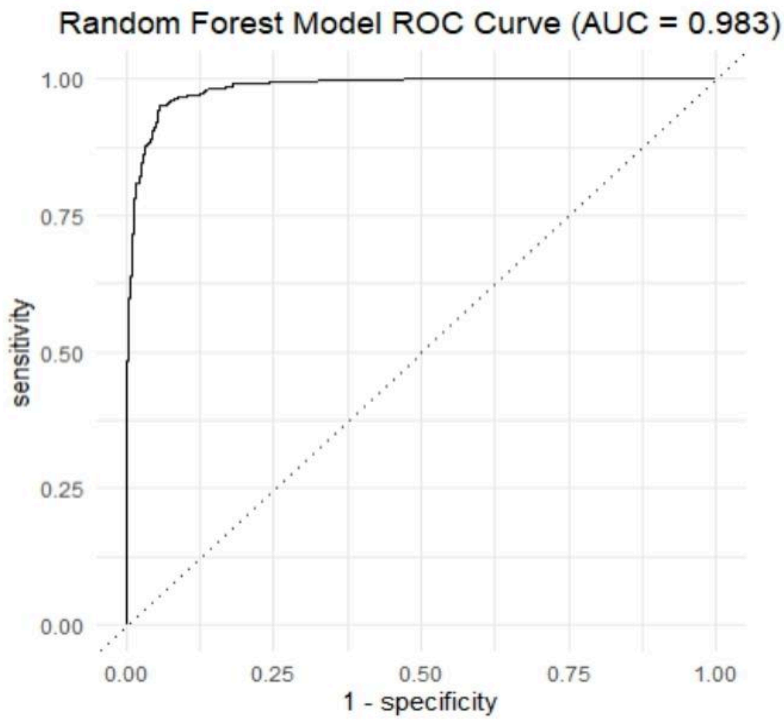

Exhibit G: ROC and AUC for XG-Boost Model

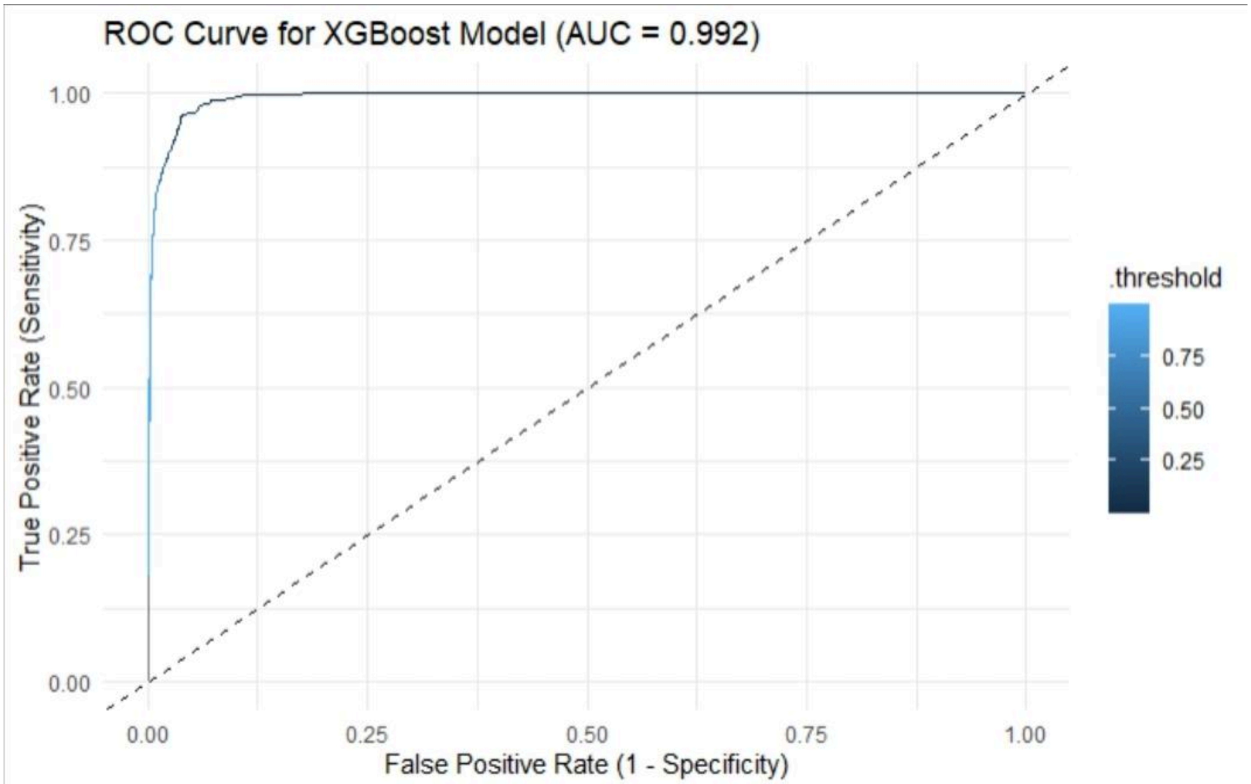

Exhibit H: Feature Importance in the XG-Boost Model